Applied Machine Learning

17-02-26

- Artificial Intelligence
- Data Science
- Machine Learning
- Deep Learning
- NLP, computer vision
- Generative AI

Artificial Intelligence

Natural intelligence - living organisms
↓
they can observe, learn
react, make decisions, move, etc.

AI means computer system, software which can mimic human behaviour.

eg.

chatgpt / claude / Gemini - chatbot - give ans to our questions
deepseek

cursor, windsurf, copilot - coding assistants
- code using natural language.

Perplexity - AI search engine

Alexa, siri — voice based chatbots

self driving cars

google translate - NLP

spotify, youtube, instagram, etc → recommendation engine
ecommerce

face recognition

Machines
computers
Appliences
Software
AI

} bring Automation

Till 2010 - we had automated most of the manual labour tasks

eg. All digital devices
computers, smartphones, TV, manufacturing machines etc.

* Customer care calls
- this was completely manual for very long time.

- small automation happened later with a rule based system

this system has lot of limitations

- it only covers FAQs, doesn't cover other queries
- response time is very long
- doesn't have human touch

20 FAQs	2000 FAQs
q_1 - 01	q_1 - 01
q_2 - 02	q_2 - 02
q_3 - 11	⋮
q_4 - 12	
q_5 - 31	
⋮	
q_{20} - 97	q_{2000} - 14572

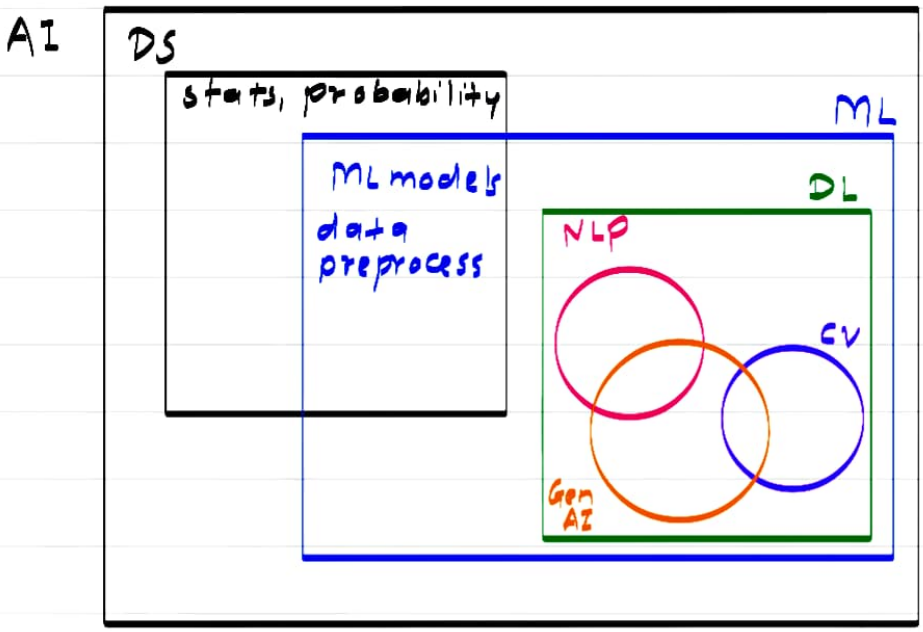
this system is not efficient for too many rules

if 0 — bank details
  if 1 - balance - q_1
  if 2 - acc statement - q_2

if 1 - card details
  if 1 - card balance
  if 2 - card bills

When huge number of rules are envolved in any automation then rule based system are not efficient but AI based systems are efficient for this.

Rule based automation	AI based automation
washing machine	self driving cars
petrol car	AI chatbots
speaker, fan	Alexa, siri



All the above technologies are part of AI and strongly connected to each other.

Data Science

- technologies, science where data plays crutial role

data — Analysis, insights, business decisions } statistics ✓

data — predicting future → machine learning

Rule based system → explicitely programmed system

↓

they have fixed behaviour

↓

fixed input and fixed output

↓

code it using programming

↓

no artificial intelligence

↓

this is automation without AI

Machine Learning

house area
pin code
rooms

→ ML model → house price

ML is a technique of teaching machine to perform any task automatically without being explicitely programmed.

ML based system → inputs and their respective outputs are not fixed.

ML systems are softwares, programs which give some output for given input.

eg. similar to a function

$$y = f(x)$$

output = f(input)

input | output

user query → [chatgpt] → Answer

face image → [face recognition] → yes / no

yes → unlock

no → stay locked

eng text → [google translate] → telugu, tamil text

ML involves creation of algorithms which have predictive capabilities

A good ML system makes accurate prediction.

Every output of an ML model is an estimated prediction.

eg. house price prediction

Hyd city → buy a 2BHK apt in locality A
↓
hitech city

Research - online - 99 acres
MB
housing

- agents

} Analyse multiple properties
↓
estimate approximate price for your apt with required features

We can automate this house price estimation process using ML.

house area
rooms
floor
city
pincode
amenities
etc

} → [ML system model] → predicted price

≈ 75 lacs